Execution-Level Correction for Surgical Robot Automation via Residual Policy Learning from Human Interventions

Kejian Shi, Qingbiao Li†, Chunhei Lau, Tianyi Xia, Lei Song, Yonghao Long, Sethu Vijayakumar, Qi Dou†

***Abstract*—Automation in robotic-assisted minimally invasive surgery (RMIS) demands millimeter-level execution precision. While imitation learning (IL) effectively captures long-horizon task structure, it falls short at deployment time in detecting and correcting localized execution drift at precision-critical stages such as grasping, handover, and tool-object contact. To address this, we present SuRResT (Surgical Robot Residual Transformer), a framework that learns execution-level corrections from human-in-the-loop (HIL) interventions collected during base policy rollouts. To extend correction coverage beyond explicit intervention timesteps, we augment the correction dataset through spatially-guided label propagation and a trajectory-aware deviation signal. A residual module is then trained on this dataset to selectively correct the frozen base policy at precision-critical states during deployment. We evaluate SuRResT across three surgical manipulation tasks in simulation and on the da Vinci Research Kit (dVRK) platform. Through targeted interventions concentrated at failure-prone states, SuRResT improves task success by up to 59.3 percentage points over the base IL policy in simulation. On the real-world dVRK hardware, SuRResT achieves 97.8% success on Needle Grasp and Place and 93.3% on Peg Transfer, substantially outperforming end-to-end retraining, finetuning, and data scaling baselines under matched human data-collection budgets.**

***Index Terms*—Surgical Robotics: Laparoscopy, Computer Vision for Automation**

I. Introduction

ROBOTIC-assisted minimally invasive surgery (RMIS) has become a cornerstone of modern healthcare, with the da Vinci surgical system serving as the most widely adopted platform worldwide [1], [2]. With growing interest in semi-autonomous and autonomous assistance [3]–[5], data-driven approaches such as imitation learning (IL) and reinforcement learning (RL) have emerged as promising tools for acquiring surgical visuomotor skills from expert demonstrations.

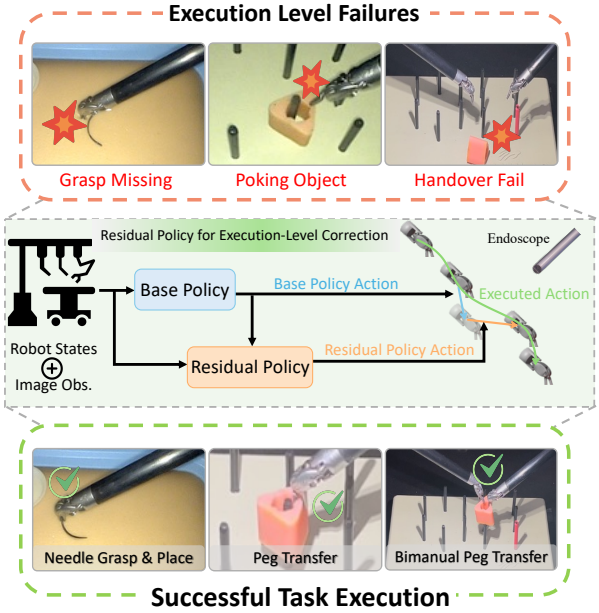


Fig. 1. SuRResT augments a base policy with localized residual corrections to mitigate execution-level failures during precision-critical surgical interactions.

Recent learning-based systems have demonstrated strong capability in learning long-horizon task structure and multi-stage manipulation skills for surgical robot automation [6]–[9]. In practice, fine-grained tasks expose a different failure mode: the policy may follow the intended sequence of actions yet still fail when small execution errors accumulate during precision-critical interactions [10], [11]. In surgical settings, instruments operate on a scale of millimeters, where small misalignment can propagate through subsequent interactions, leading to unintended contact and potential task failure. This motivates correcting emerging execution deviations during ongoing manipulation, before they result in observable failures.

A natural response is to collect more demonstrations or retrain the policy end-to-end [12], [13]. However, localized execution errors may remain even with larger demonstration datasets [14]. This suggests that the underlying cause is not a lack of task coverage, but the absence of dedicated execution-level correction mechanisms. A more principled approach is to learn a residual correction on top of a frozen base policy. Residual refinement has shown promise for improving precision in robotics [14], [15], and recent work demonstrates that residual policies trained from human-in-the-loop (HIL)

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corrections can substantially improve performance in contact-rich manipulation [16]. However, these approaches rely on online RL with trial-and-error, force feedback, or multi-view sensing, none of which are available in endoscopic surgical settings at this stage.

In this work, we adapt this residual correction paradigm to surgical robotics and present SuRResT (Surgical Robot Residual Transformer). The key insight is that human operators naturally possess the precision and contextual awareness to provide targeted corrections at exactly the moments and states where the base policy requires intervention, as shown in Fig. 1. By collecting such expert corrections during base policy rollouts and learning a residual module from them, we can directly supervise the failure-prone states that are difficult to address through standard data scaling alone. Since human operators intervene only when failures become imminent, the collected corrections are inherently sparse. To extend corrective supervision beyond these explicit intervention timesteps, we augment the correction dataset through an annotation-free strategy that covers pre-failure and off-trajectory states. Unlike prior residual approaches that rely on force feedback or multi-view sensing, SuRResT operates purely from single-view endoscopic vision and robot state, making it compatible with standard endoscopic surgical robot sensing configurations.

We evaluate the proposed framework on three surgical manipulation tasks in simulation and on the da Vinci Research Kit (dVRK) platform. SuRResT consistently outperforms baseline approaches based on end-to-end retraining and finetuning across all studied tasks, in both simulation and real-world dVRK hardware. Importantly, ablation studies further demonstrate that the IL policy trained on scaled expert demonstrations exhibits saturating performance gains, whereas SuRResT achieves substantially higher task success under the same number of demonstrations and human-corrected rollouts. To our knowledge, this is the first work to address execution-level correction with residual policy in surgical robotics. The main contributions of this work are as follows:

- **SuRResT Framework:** We present a decoupled residual framework that augments a frozen IL policy with execution-level correction.
- **Correction Data Construction:** A practical correction data construction pipeline for surgical robot deployment that extends sparse HIL interventions into structured execution-level supervision through spatially-guided label propagation and a trajectory-aware deviation signal.
- **Experimental Validation:** We validate SuRResT across simulation and real-world dVRK hardware, outperforming global policy update strategies under matched human-supervision budgets.

II. Related Work

A. Surgical Robot Automation

Early efforts in surgical robot autonomy demonstrated the feasibility of automating surgical actions such as suturing [8], [17], endoscope control [18], soft tissue manipulation [19], and more complex procedural steps including intestinal anastomosis [9]. However, these approaches often rely on task-specific engineering and predefined models, limiting their adaptability and scalability.

Recent data-driven methods use imitation learning (IL) to improve generalizability across surgical tasks. Kim et al. [20] combine deep IL with optimal control for autonomous retinal surgery, while Li et al. [21] propose a 3D perception-driven IL framework for laparoscope control under limited demonstrations. More recently, SRT [12] and SRT-H [13] have demonstrated IL-based automation of tasks including needle driving, knot tying, and vessel clipping. SutureBot [7] further establishes autonomous suturing as a benchmark for dexterous IL on the dVRK. Nevertheless, existing methods remain sensitive to compounding precision errors, motivating the execution-level residual correction studied in this work.

B. Deployment-time Policy Refinement and Failure Handling

Fine-grained manipulation poses a significant challenge for imitation-learned policies, as subtle errors can compound over time [10], [11] and lead to task failure. To address this, recent work has explored residual learning, a paradigm in which a secondary policy is trained to augment a base policy with targeted corrections, thus improving precision and robustness in complex interaction tasks. For instance, Ankile et al. [14] propose a residual reinforcement learning method on top of an IL policy for industrial assembly tasks. Zhao et al. [22] integrate haptic feedback with visual priors to refine manipulation skills through real-world reinforcement learning, enabling generalizable policies in contact-rich scenarios. Jiang et al. [23] introduce TRANSIC, a sim-to-real pipeline that incorporates online human corrections to iteratively improve policy performance after deployment. Similarly, Xu et al. [16] use a compliant residual DAgger framework with HIL supervision, where correction and policy refinement are supported by force feedback and compliant intervention hardware. Beyond directly refining policy actions, complementary work addresses deployment-time failures through monitoring and recovery. Runtime-monitoring methods detect impending failures through policy consistency, uncertainty, distribution shift, or task-progress assessment [24]–[27], but do not directly specify the continuous pose adjustment required to complete an ongoing precision-critical interaction. Recovery approaches instead use language-guided policies or explicit object-centric representations to return from failed or out-of-distribution states [28], [29]. In contrast to these monitoring and recovery settings, SuRResT performs pre-failure execution-level correction under single-view endoscopic sensing without force feedback, while preserving the frozen base policy.

III. Method

A. Overview

The SuRResT pipeline is illustrated in Fig. 2 and consists of three stages. (1) **Base policy training:** a long-horizon visuomotor policy is trained by behavior cloning and then frozen for deployment (Sec. III-B). (2) **HIL Correction Collection and Augmentation:** HIL interventions are collected during base-policy rollout to form the correction dataset (Sec. III-C), and augmented with additional correction supervision for deviated

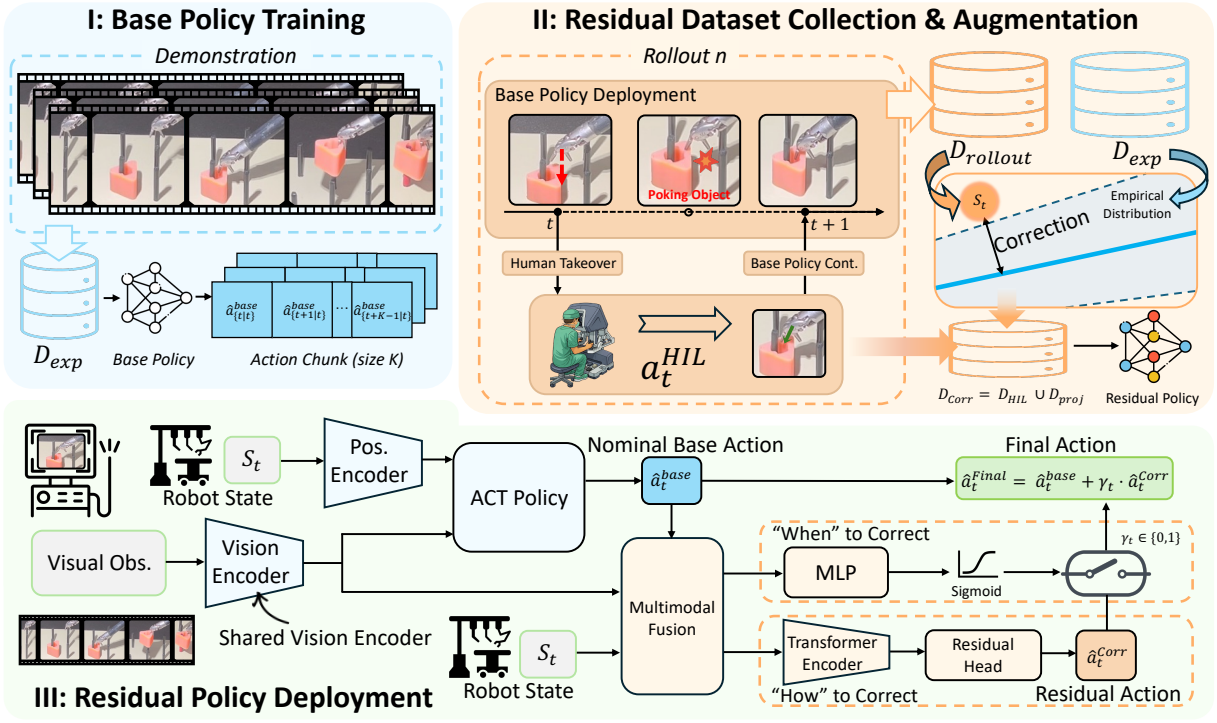


Fig. 2. SuRResT comprises three stages: (I) training and freezing an IL base policy; (II) constructing the correction dataset from HIL interventions and deviation-based labels; and (III) predicting when and how to correct the base action using a transformer-based residual policy.

states via a trajectory-aware deviation signal (Sec. III-D). (3) **Correction learning:** a residual model is trained to determine both when correction is needed and how the corresponding corrective action should be applied (Sec. III-E).

B. Training the Base Policy via Imitation Learning

We first train a visuomotor imitation policy to model the task progression. We adopt the Action Chunking Transformer (ACT) [30] as the base policy due to its receding-horizon execution and ability to predict short-horizon action sequences.

For each task, we collect a demonstration dataset $D_0 = \{\tau_i\}_{i=1}^{N}$ of expert trajectories recorded at 10 Hz, where each trajectory contains tuples (o_t, s_t, a_t). Here, o_t denotes the endoscopic image, $s_t = (p_t, q_t, g_t)$ denotes the 8-DoF robot state of a single arm under the camera frame, with $p_t \in \mathbb{R}^3$ the end-effector position, $q_t \in \mathbb{H}$ the unit quaternion orientation, and $g_t \in \mathbb{R}$ the gripper aperture. Actions are parameterized as $a_t = (\Delta p_t, \Delta q_t, g_t)$, where $(\Delta p_t, \Delta q_t)$ denote relative pose updates and g_t specifies the target gripper value.

ACT predicts an action chunk of length K,

$$\hat{\mathbf{a}}_t = [\hat{a}_{t|t}, \hat{a}_{t+1|t}, \ldots, \hat{a}_{t+K-1|t}] \sim \pi^{IL}(o_t, s_t),$$

where $\hat{a}_{t+i|t}$ denotes the prediction for timestep $t+i$ made at timestep t. Predicting short-horizon action chunks helps mitigate compounding errors in IL.

Let π_θ^{IL} denote the ACT policy parameterized by θ. The base policy is trained by behavior cloning to maximize the likelihood of demonstrated action chunks:

$$\theta^\star = \arg\min_\theta - \sum_{(o_t, s_t, \mathbf{a}_t) \in D_0} \log \pi_\theta^{IL}(\mathbf{a}_t \mid o_t, s_t). \quad (1)$$

After training, the optimized policy $\pi^{IL} := \pi_{\theta^\star}^{IL}$ is frozen and used as the base policy in all subsequent stages.

During deployment, overlapping predictions are combined through weighted temporal ensembling:

$$\hat{a}_t^{\mathrm{IL}} = \sum_{i \in P_t} w_{t,i} \hat{a}_{t|i}, \qquad \sum_{i \in P_t} w_{t,i} = 1, \quad (2)$$

where P_t denotes the prediction set for timestep t, and $w_{t,i}$ weights the prediction made at timestep i.

C. Human-in-the-Loop Correction Data Collection

The frozen base policy π^{IL} is deployed under expert supervision to collect correction labels. At each timestep t, the policy executes the nominal action $\hat{a}_t^{IL}$ based on the current situation. The operator intervenes only when the robot approaches failure at a precision-critical step, such as a misaligned grasp or unstable handover. In such cases, control is temporarily transferred to the operator via teleoperation to restore the robot to the operator-corrected end-effector configuration, after which execution resumes under the base policy.

An intervention at timestep t defines a corrective action

$$a_t^{Corr} = (\Delta p_t^{Corr}, \Delta q_t^{Corr}, g_t^{Corr}), \quad (3)$$

where $(\Delta p_t^{Corr}, \Delta q_t^{Corr})$ denote the relative pose adjustment applied by the operator and g_t^{Corr} is the corresponding gripper target. For bimanual tasks, the state and action representations are formed by concatenating the corresponding vectors of the two manipulators. For each timestep, we store a tuple $(o_t, s_t, \hat{a}_t^{IL}, a_t^{Corr}, y_t)$, where $y_t \in \{0, 1\}$ indicates whether a correction is defined. The intervention timesteps are labeled positive ($y_t = 1$), while all others are labeled negative ($y_t = 0$)

with $a_t^{Corr} = 0$. The collection of all such tuples forms the correction dataset D_{corr}.

Because interventions are typically triggered only when failure becomes imminent, labeling only the intervention timestep may miss the earlier drift that contributes to the failure. To provide supervision for these pre-failure states, we propagate each intervention label to preceding timesteps whose end-effector position lies within a spatial neighborhood of the intervention state, using a radius equal to the task-specific mean translational action magnitude across the demonstration trajectories, ensuring that the propagation radius is consistent with the scale of robot execution. For each such timestep, we define the correction target as the movement from the local state to the corrected state reached by the operator. All propagated labels correspond to the same corrective outcome while remaining conditioned on the local state at each timestep. A sensitivity study of radii choices is conducted in IV-D.

D. Deviation Signal for Correction Dataset Augmentation

Beyond propagating labels near observed interventions, a second source of unlabeled states arises when the robot drifts from the demonstrated manifold without triggering operator takeover. Although these states may not yet exhibit visible failure, their deviation from the demonstrated manifold makes them correction relevant, as uncorrected drift can propagate into precision-critical failures.

Reference Manifold Construction. From D_0, we construct a reference manifold by resampling all demonstration trajectories to a common length T, set to the mean trajectory length across all demonstrations, and computing a mean trajectory $\bar{p}(t)$ with per-timestep standard deviation $\sigma(t)$:

$$\bar{p}(t) = \frac{1}{N}\sum_{i=1}^{N} p_i(t), \quad \sigma(t) = \sqrt{\frac{1}{N}\sum_{i=1}^{N} \|p_i(t) - \bar{p}(t)\|_2^2} \tag{4}$$

This yields a tolerance envelope that captures the natural variation across expert demonstrations and defines the boundary of the demonstrated execution manifold.

Deviation Detection. For each rollout timestep t, let $p_t \in \mathbb{R}^3$ denote the current end-effector position. During offline dataset augmentation, we identify the nearest spatial correspondence on the mean demonstrated trajectory and compute the corresponding deviation

$$\tau^* = \arg\min_{\tau} \|\bar{p}(\tau) - p_t\|_2, \quad d_t = \|\bar{p}(\tau^*) - p_t\|_2 \tag{5}$$

and mark $y_t = 1$ if $d_t > \sigma(\tau^*)$, and $y_t = 0$ otherwise. This criterion identifies states where the robot has drifted beyond the natural execution variability of expert demonstrations, making them candidates for correction supervision.

Trajectory-Aware Correction Target Assignment. For each timestep with $y_t = 1$, we use τ^* to define a normalized trajectory position and retrieve the corresponding state from each demonstration in D_0. The correction target is selected as the spatially nearest among these trajectory-indexed candidates, rather than by searching across all demonstration waypoints, thereby reducing cross-phase target assignment.

In our evaluated tasks, different stages can overlap spatially, but such overlap mainly occurs between adjacent stages with similar corrective directions, yielding similar return-to-manifold targets. We retrieve the corresponding demonstration state $s^*(\tau^*) = (p^*(\tau^*), q^*(\tau^*), g^*(\tau^*))$ and compute the correction action as the movement from the current state back to the demonstrated manifold:

$$a_t^{\text{Corr}} = \left(p^*(\tau^*) - p_t,\ q^*(\tau^*) \otimes q_t^{-1},\ g^*(\tau^*)\right), \tag{6}$$

where the position and rotation components are expressed as relative displacements and the gripper target is specified as an absolute value, consistent with the action parameterization in Sec. III-B. This provides broad geometric recovery supervision, while direct HIL interventions provide precise corrections at critical manipulation stages.

The labeled tuple $(o_t, s_t, \hat{a}_t^{IL}, a_t^{\text{Corr}}, y_t)$ is appended to D_{corr}. Together with the intervention label propagation in Sec. III-C, this augmentation strategy ensures that D_{corr} provides expanded geometric correction supervision across both observed failure states and latent drift states, without requiring any additional human annotation beyond the corrections already collected during deployment.

E. Surgical Robot Residual Transformer

The correction model takes as input the visual observation o_t, the current robot state s_t, and the predicted action $\hat{a}_t^{IL}$ from the base policy. We reuse the frozen ACT visual encoder to extract features from o_t, and fuse them with embeddings of s_t and $\hat{a}_t^{IL}$ to form a shared multimodal representation

$$h_t = f_\phi(o_t, s_t, \hat{a}_t^{IL}). \tag{7}$$

On top of this representation, the network uses a dual-branch design that predicts both *how* to correct by Eq. 8 and *when* correction is needed by Eq. 10.

The correction branch models the corrective action:

$$\hat{a}_t^{Corr} = \mu_\theta(h_t), \tag{8}$$

and is trained on correction-relevant samples using

$$\mathcal{L}_{Corr} = \mathbb{E}_{(o_t, s_t, \hat{a}_t^{IL}, a_t^{Corr}, y_t) \sim D_{\text{corr}},\, y_t=1} \left[\left\| \hat{a}_t^{Corr} - a_t^{Corr} \right\|_2^2 \right]. \tag{9}$$

The activation branch predicts whether correction should be applied:

$$\gamma_t = Binary(g_\psi(h_t)), \qquad \gamma_t \in \{0, 1\}, \tag{10}$$

and is trained on all samples in D_{corr} using the binary cross-entropy loss

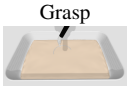
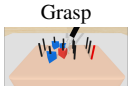
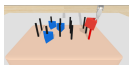
$$\begin{aligned} \mathcal{L}_{gate} = \mathbb{E}_{(o_t, s_t, \hat{a}_t^{IL}, y_t) \sim D_{\text{corr}}} \Big[& - y_t \log g_\psi(h_t) \\ & - (1 - y_t) \log(1 - g_\psi(h_t)) \Big] \end{aligned} \tag{11}$$

The two branches play complementary roles: the correction branch predicts the corrective action, while the activation branch determines when that action should be applied. The final executed action is

$$\hat{a}_t^{Final} = \hat{a}_t^{IL} + \gamma_t \hat{a}_t^{Corr}, \tag{12}$$

so that the residual module acts as a selective execution-level corrector on top of the frozen base policy. We conduct simulation experiments to demonstrate that the proposed method

TABLE I
PER-PHASE SUCCESS RATES (%) ACROSS THREE TASKS IN SIMULATION, COMPARING SURRESST WITH BASELINES.

Method	Needle Grasp	Peg Transfer		Bimanual Peg Transfer		
	Grasp	Grasp	Place	Grasp	Handover	Place
						
ACT_{D_0}	60.7±0.5	64.3±4.2	37.0±1.6	100.0±0.0	61.3±0.9	34.0±1.4
$ACT_{D_0^+}$	94.7±2.9	92.7±1.7	76.3±1.7	100.0±0.0	**99.3±0.5**	60.0±2.9
$ACT_{finetune}$	78.3±1.3	48.3±3.7	28.3±1.3	85.7±3.4	44.0±4.7	16.7±2.1
DAgger-1R	83.0±5.1	84.0±2.2	63.0±2.9	98.7±0.9	86.0±2.8	53.7±6.2
SuRResT (Ours)	**99.7±0.5**	**97.3±0.5**	**96.3±1.7**	**100.0±0.0**	96.0±0.0	**91.7±1.7**

correctly predicts when and how to correct, as analyzed in Sec. IV-B (Fig. 3). The residual is applied to the base action executed at the current timestep. When the gating branch activates, predictions generated from pre-correction states are discarded. At the next timestep, the base policy generates a new action chunk from the corrected observation and robot state. When no correction is activated, the original ACT procedure is retained. We do not dynamically modify the action-chunk horizon.

F. Implementation Details

All methods were trained for 5,000 epochs without early stopping or a validation split on a single NVIDIA RTX 3090, and the checkpoint with the lowest training loss was selected for evaluation. ACT-based policies used an action-chunk horizon of $K = 5$. For correction-derived samples used to train the ACT-based baselines, supervision was applied only to the first action in each chunk, while the remaining chunk positions were masked from the loss. For Needle Grasp and Place, Peg Transfer, and Bimanual Peg Transfer, the HIL propagation radii were the task-specific mean translational action magnitudes: 2.61, 2.22, and 2.02 mm in simulation, and 1.64, 1.15, and 0.97 mm in the real world. We let D_{HIL} denote positive states obtained from direct interventions and spatial propagation, whereas D_{proj} comprises deduplicated deviation-derived positives; all remaining rollout states are negative gate labels. Macro-averaged across tasks, $D_{\text{HIL}}/D_{\text{proj}}$ represented $2.3\%/1.2\%$ of simulation states and $4.7\%/6.6\%$ of hardware states, yielding positive-to-negative ratios of approximately 1.0:27.2 and 1.0:7.8, respectively.

IV. EXPERIMENTS

A. Evaluation Setup

We evaluate SuRResT on three representative manipulation tasks from the da Vinci surgical skills curriculum [31]: Needle Grasp and Place, Peg Transfer, and Bimanual Peg Transfer. For the Needle Grasp and Place task, in the simulator we evaluate only the Needle Grasp phase, whereas physical deployments test the full Needle Grasp and Place sequence. These tasks jointly cover fine-grained motor control, spatial coordination, and bimanual dexterity. To assess robustness, comparable task-specific randomization was applied in both simulation and real-world evaluation. For Needle Grasp and Place, the initial needle position was sampled within a 4.5×3.0 cm elliptical region, with its orientation perturbed by up to $\pm 5^\circ$. For Peg Transfer and Bimanual Peg Transfer, the pegboard center was displaced by up to ± 1.0 cm along the x- and y-axes, and the block orientation was perturbed by up to $\pm 5^\circ$. Experiments are conducted both in the SurRoL simulation environment [32] and on dVRK, using only a single endoscopic camera view to reflect clinically realistic sensing constraints.

We compare SuRResT against four baselines:

- **Base Policy Only (ACT_{D_0}):** the IL policy π^{IL} trained on a dataset D_0 of 50 expert trajectories.
- **Single-round DAgger (DAgger-1R):** a DAgger-style baseline retrained from scratch on $D_0 \cup D_{\text{Corr}}$ using corrective labels collected from additional policy rollouts.
- **Finetuned Policy (ACT_{finetune}):** the base policy π^{IL} further finetuned using the correction dataset D_{Corr}.
- **More Demonstrations ($ACT_{D_{0+}}$):** an IL policy trained with the same number of additional human-collected expert trajectories as the number of correction rollouts used by the correction-based methods.

These baselines represent three standard ways of exploiting additional supervision: direct finetuning, dataset aggregation with end-to-end retraining, and scaling demonstration data. This comparison allows us to isolate whether the performance gain of SuRResT comes from the correction data itself or from the residual correction formulation used to incorporate it.

B. Simulation Results

We first evaluate SuRResT in simulation using SurRoL [32], which enables controlled analysis of corrective behavior and task-level performance. For each task, the base IL policy π^{IL} is trained on 50 expert demonstrations, while the residual correction policy π^{Corr} is trained on an augmented correction dataset constructed from 50 base policy rollouts. All methods are evaluated over 100 rollouts per task across three random seeds, and results are reported as mean $\pm$ standard deviation.

As shown in Tab. I, SuRResT achieves the highest final task completion rates across all three simulation tasks, demonstrating strong performance in single-arm manipulation and reliable execution in bimanual coordination. Per-phase evaluation reveals that while the base ACT policy maintains high-level task progression, it frequently fails at the grasping and handover phases where execution precision is most critical, whereas SuRResT mitigates these bottlenecks by providing targeted refinements. Importantly, both ACT_{finetune} and *DAgger-1R* yield only modest improvements over the base policy, despite using the same correction dataset. This

TABLE II

PER-PHASE SUCCESS RATES (%), REPORTED AS MEAN ± STANDARD DEVIATION ACROSS THREE INDEPENDENT SESSIONS OF 30 TRIALS EACH. THE 95% CI COLUMNS REPORT EXACT TWO-SIDED CLOPPER–PEARSON INTERVALS FOR POOLED WHOLE-TASK SUCCESS OVER 90 TRIALS.

Method	Needle GP			Peg Trans.			Bi-Peg Trans.			
	G	Pl	95% CI	G	Pl	95% CI	G	H	Pl	95% CI
ACT_{D_0}	51.1 ±6.9	44.4 ±1.9	[34.0, 55.3]	48.9 ±5.1	7.8 ±6.9	[3.2, 15.4]	36.7 ±3.3	3.3 ±3.3	1.1 ±1.9	[0.0, 6.0]
$ACT_{D_0^+}$	73.3 ±3.3	65.6 ±5.1	[54.8, 75.3]	75.6 ±5.1	15.6 ±6.9	[8.8, 24.7]	81.1 ±12.6	6.7 ±0.0	3.3 ±3.3	[0.7, 9.4]
ACT_{ft}	58.9 ±1.9	48.9 ±5.1	[38.2, 59.7]	66.7 ±10.0	14.4 ±11.7	[7.9, 23.4]	62.2 ±10.7	2.2 ±3.8	1.1 ±1.9	[0.0, 6.0]
DAgger-1R	71.1 ±10.7	65.6 ±5.1	[54.8, 75.3]	68.9 ±9.6	16.7 ±6.7	[9.6, 26.0]	64.4 ±10.2	13.3 ±8.8	5.6 ±6.9	[1.8, 12.5]
SuRResT	**97.8 ±3.8**	**97.8 ±3.8**	**[92.2, 99.7]**	**98.9 ±1.9**	**93.3 ±3.3**	**[86.1, 97.5]**	**96.7 ±3.3**	**74.4 ±10.7**	**65.6 ±10.7**	**[54.8, 75.3]**

G: Grasp; Pl: Place; H: Handover. Confidence intervals correspond to whole-task completion, i.e., the final phase of each task.

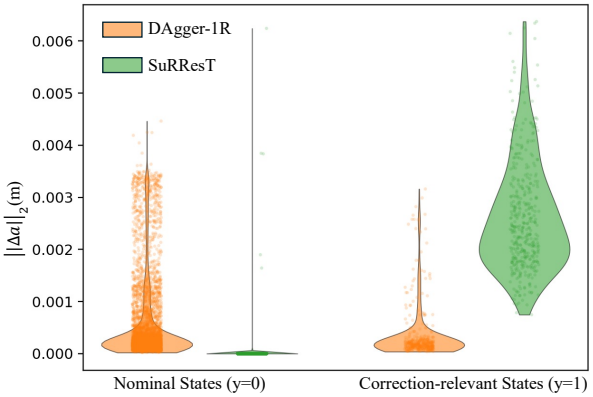


Fig. 3. Distribution of action-change magnitudes relative to the frozen base policy for DAgger-1R and SuRResT on Peg Transfer, evaluated over nominal and correction-relevant states. Violin plots show the distributions, and overlaid scatter points denote individual evaluation states.

suggests that simply injecting corrective supervision into end-to-end policy updates is insufficient to fully exploit its value. By contrast, SuRResT incorporates these signals through its dedicated residual correction module, achieving substantially higher task success.

To further characterize how the same correction supervision is absorbed under different training schemes, we analyze the changes in predicted actions induced relative to the base policy on a held-out dataset of 20 Peg Transfer rollouts collected under the same deployment conditions. The dataset is partitioned into nominal execution states ($y = 0$) and correction-relevant states ($y = 1$) following Sec. III-C and III-D. We compute the translational action-change magnitude as $\|\Delta\hat{a}\|_2 = \|\hat{a}^{\pi}(s) - \hat{a}^{\text{base}}(s)\|_2$, where $\hat{a}^{\pi}(s)$ denotes the translational component of the action predicted at state s by either $DAgger\text{-}1R$ or $SuRResT$, and $\hat{a}^{\text{base}}(s)$ denotes the corresponding prediction of the frozen base policy.

As shown in Fig. 3, the two methods induce markedly different prediction-shift distributions. On nominal execution states, end-to-end retraining exhibits a broad distribution of action changes with relatively small magnitudes, indicating that the learned update is distributed across a wide range of states. On correction-relevant states, SuRResT produces substantially larger action shifts, reflecting targeted corrections, whereas DAgger-1R remains concentrated at smaller magnitudes. The gating branch achieves a precision of 0.988, a recall of 1.000, and an F1-score of 0.994. Overall, the simulation results show that the benefit of SuRResT does not arise merely from access to additional data, but from how that corrective information is structured and deployed. Notably, SuRResT achieves these high success rates while the gating branch activates the residual module on only 2.7% of rollout timesteps for Needle Grasp, 6.1% for Peg Transfer, and 7.4% for Bimanual Peg Transfer. This combination of performance gains and high sparsity confirms that the vast majority of execution states do not require adjustment.

C. Real-World Experiments on dVRK

We further assess SuRResT on the dVRK platform to examine whether the advantages observed in simulation persist under real-world robot execution. Compared with simulation, hardware deployment introduces additional challenges, including image noise, calibration inaccuracy, actuation uncertainty, and contact-rich interaction dynamics. Our experimental setup and task randomization ranges are illustrated in Fig. 4. For each task, the base imitation policy is trained from 50 teleoperated demonstrations collected using the Master Tool Manipulators, and the residual correction policy is learned from 30 corrected rollouts obtained through human intervention and deviation-based augmentation. Physical evaluation was conducted across three independent sessions on separate days, with 30 trials per task in each session, and results are reported as mean ± standard deviation across sessions. The endoscope pose and arm configurations were fixed, and calibration accuracy was verified before each session without recomputing the robot-to-camera transformation. To ensure safe physical interaction during real-world deployment, the final executed actions are subject to a strict velocity clamp, preventing rapid movements that could lead to instrument damage or uncontrolled collisions with the environment.

Fig. 4 further illustrates the role of the residual correction module during physical execution. During physical deployment, unsuccessful trials typically stem from gradual end-effector drift at the same precision-critical interaction stages identified in simulation. In these situations, SuRResT applies small action adjustments that restore alignment before the deviation develops into a missed grasp, unstable transfer, or task-level failure.

The quantitative results in Tab. II show that SuRResT retains a clear advantage on real hardware. It achieves 97.8% success on Needle Grasp and Place, 93.3% on Peg Transfer, and 65.6% on Bimanual Peg Transfer, with corresponding exact 95% binomial confidence intervals of $[92.2, 99.7]$, $[86.1, 97.5]$, and $[54.8, 75.3]$, respectively. This trend is most evident in the full-task metrics, where methods based on end-to-end policy updating remain noticeably less reliable even when provided with the same corrective experience. The lower success

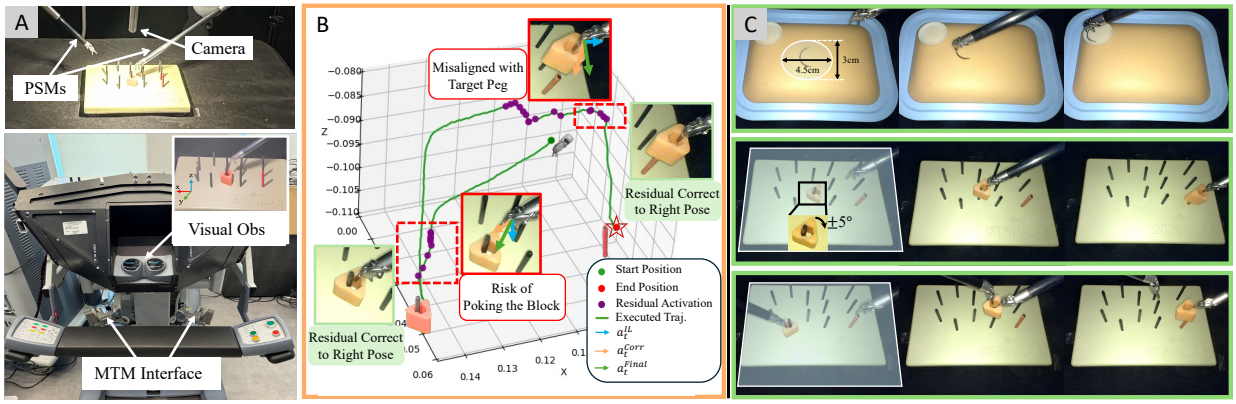


Fig. 4. (A) Real-world dVRK setup. (B) Example SuRResT trajectory on Peg Transfer, where the residual policy activates upon execution drift and realigns the end-effector with the target. (C) Successful executions across tasks, with the corresponding task-randomization ranges illustrated.

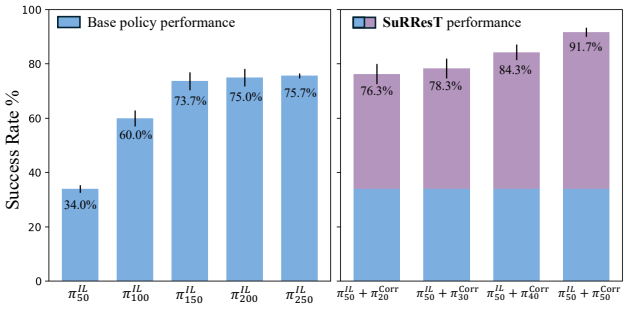


Fig. 5. Performance on simulated Bimanual Peg Transfer with increasing numbers of demonstrations and correction trajectories. (left): IL policy with scaling demonstrations D_0 from 50 to 250. (right): SuRResT performance with same π^{IL}_{50} base IL policy and π^{Corr} with scaling correction dataset D_{Corr} from 20 to 50.

rate on Bimanual Peg Transfer reflects compounded inter-arm coordination errors. SuRResT corrects localized action deviations but inherits the coordination strategy of the frozen base policy; it cannot introduce synchronized motion, role allocation, or handover timing that the base policy has not learned. Consequently, a failed handover also prevents the placement phase from being attempted.

D. Ablation Study

1) Comparison with Additional Demonstrations: We first compare SuRResT against the alternative of increasing the number of expert demonstrations in simulation. As shown in Fig. 5 (left), IL performance improves with additional demonstrations but gradually saturates. We then compare this trend with SuRResT trained using 50 imitation demonstrations together with 50 correction trajectories. Under this setting, SuRResT achieves 91.7% success, whereas an IL policy trained on 250 demonstrations achieves 75.7%. This result indicates that SuRResT benefits from correction trajectories targeting the execution errors of the deployed policy, rather than merely from additional expert data. Fig. 5 (right) further shows the effect of varying the number of correction trajectories. Performance improves as more correction trajectories are added, indicating that the correction data contributes directly to the final task performance.

2) Sensitivity to the Propagation Radius: We evaluate the propagation radius on Bimanual Peg Transfer in simulation over three independently trained runs. Radii of $0.2\times$, $1\times$, $2\times$, and $10\times$ the default value yielded propagated HIL-label rates of 0.0%, 2.1%, 4.2%, and 15.9% of rollout states, respectively, with corresponding success rates of $81.7\pm4.0\%$, $91.7\pm1.7\%$, $86.3\pm2.5\%$, and $74.0\pm4.4\%$. The default radius, defined as the mean translational action magnitude ($r = 2.02$ mm), provides the best balance between label coverage and locality.

TABLE III
PER-PHASE SUCCESS COUNTS FOR THE ABLATION VARIANTS.

Variant	Needle GP		Peg Trans.		Bi-Peg Trans.		
	G	Pl	G	Pl	G	H	Pl
w/o HIL	16	14	14	13	20	6	0
w/o Dev.-Aug.	28	27	29	20	28	16	12
w/o Gating	17	15	16	14	18	1	1
Full	**30**	**30**	**30**	**29**	**28**	**21**	**21**

G: Grasp, Pl: Place, H: Handover. All counts are out of 30.

3) Contribution of Each SuRResT Component: We examine the contribution of the main components of SuRResT on the physical dVRK across all three tasks. Each ablation variant was evaluated in a separate set of 30 physical trials. Specifically, we evaluate ablated variants to isolate the contributions of the key components. The results are summarized in Tab. III. Together, these results confirm the complementary contributions of each component.

Removing HIL correction data leads to the largest performance drop across all tasks, indicating that direct intervention data provide the largest contribution to the final performance of SuRResT. Removing deviation-based augmentation also reduces performance, although less severely than removing HIL correction data.

The no-gating variant resembles the always-on residual formulation of CR-DAgger [16]. Removing the gating mechanism also substantially reduces performance. This effect is particularly pronounced in the bimanual task, where unintended corrections at nominal states can disrupt inter-arm coordination. This consistent reduction across tasks shows that the performance gain of SuRResT does not come solely from learning a residual correction term, but also from deciding when that correction should be applied.

V. Conclusion

In this work, we presented SuRResT, a gated residual correction framework that augments a frozen IL policy using targeted HIL corrections. By separating global task progression from localized execution-level refinement, SuRResT learns both when and how to correct precision-critical deviations without retraining the base policy. Experiments in simulation and on the physical dVRK demonstrate substantial improvements over end-to-end retraining, finetuning, and additional-demonstration baselines.

Because SuRResT uses a frozen base policy, it inherits limitations in multimodal behavior and inter-arm coordination. Its phase-matching mechanism also assumes that task progression can be inferred from the current state, which may introduce ambiguity in more complex procedures. Incorporating temporal context or progression constraints could help address this issue. Nevertheless, the results demonstrate that selective residual correction provides an effective approach to improving fine-grained surgical robot autonomy.

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